

**Assignment Report**  
**PayFlow Business Solutions**  
**Invoice Late Payment Prediction Decision Tree Report**

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## **Task 1: Understand the Business Problem**

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### **What is the business about?**

PayFlow Business Solutions is a B2B financial operations company that helps small and medium-sized businesses manage invoicing, payment collection, and accounts receivable. The company provides invoice tracking, payment reminders, online payment portals, and financial reporting dashboards for businesses that sell products or services to other companies. PayFlow works with clients in industries such as retail, professional services, construction, healthcare, manufacturing, and technology. Its customers often issue invoices with payment terms such as 15, 30, 45, or 60 days.

### **What problem is the business trying to solve?**

Late invoice payments create serious cash-flow problems for businesses. When customers pay late, companies may struggle to pay suppliers, manage payroll, invest in operations, or plan future growth. Late payment also increases administrative workload because staff must send reminders, follow up with customers, and sometimes handle disputes. PayFlow wants to identify invoices that are at high risk of being paid late before the due date passes. This would allow the company to prioritize follow-up actions and support customers earlier, reducing both financial risk and operational burden.

### **What decision can machine learning help the business make?**

Machine learning can help PayFlow predict whether an invoice is likely to be paid late. A Decision Tree classification model can use customer history, invoice characteristics, payment behaviour, and service interaction variables to classify each invoice as likely late or likely on time. The prediction can support decisions such as which invoices should receive early reminders, which customers may need account manager support, and which invoices may require review before applying collection actions.

### **What is the target variable in the dataset?**

The target variable is `Late_Payment`. It is a binary variable with two possible values: Yes, meaning the invoice is expected to be paid late, and No, meaning the invoice is expected to be paid on time. This is the variable the Decision Tree model is trained to predict.

### **What are the input features?**

The input features are `Customer_Segment`, `Industry`, `Region`, `Invoice_Month`, `Invoice_Amount`, `Payment_Terms_Days`, `Customer_Tenure_Months`, `Prior_Late_Payments`, `Avg_Days_To_Pay`, `Days_Since_Last_Order`, `Dispute_Flag`, `Reminder_Count`, `Online_Portal_Use`, `Early_Payment_Discount_Offered`, and `Credit_Score_Band`.

### **Why is this prediction useful for the business?**

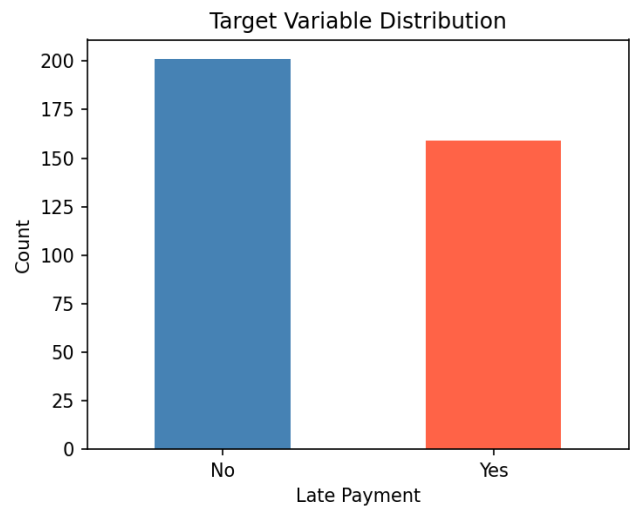
This prediction is useful because instead of treating every invoice the same way, PayFlow can focus earlier attention on invoices that are more likely to become late. This improves cash-flow planning, reduces manual collection workload, and helps businesses communicate with customers before payment issues become serious. The model can also help managers understand which factors are most associated with late payment so that targeted actions such as dispute resolution support, payment portal training, or proactive reminders can be designed for specific customer groups.

## Task 2: Data Preparation

### Dataset Overview

| Item                | Value |
|---------------------|-------|
| Total Rows          | 360   |
| Total Columns       | 17    |
| Missing Values      | 0     |
| Duplicate Rows      | 0     |
| Rows after Cleaning | 360   |
| Late Payment: No    | 201   |
| Late Payment: Yes   | 159   |

### Target Variable Distribution

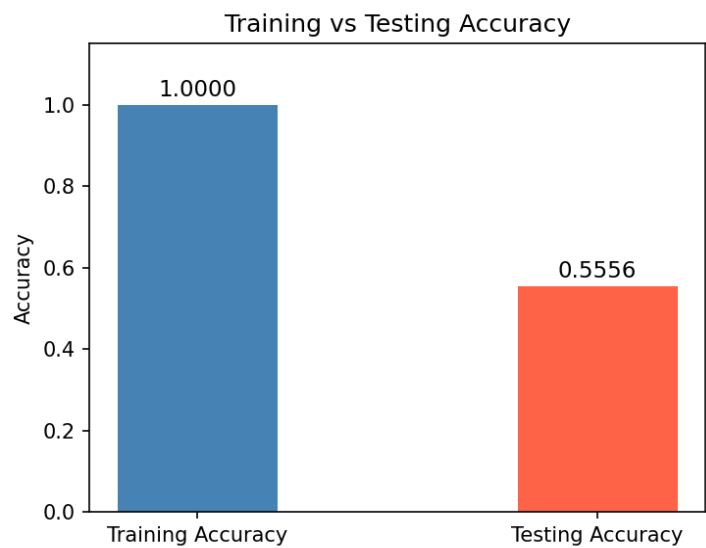


### Cleaning Steps

The dataset had no missing values and no duplicate rows, so no removal was required. Invoice\_ID was dropped as it is only an identifier and not a useful feature. Categorical features were encoded using Label Encoding. The data was split into 80% training and 20% testing sets using stratified sampling to maintain class balance.

### Task 3: Decision Tree Model Results

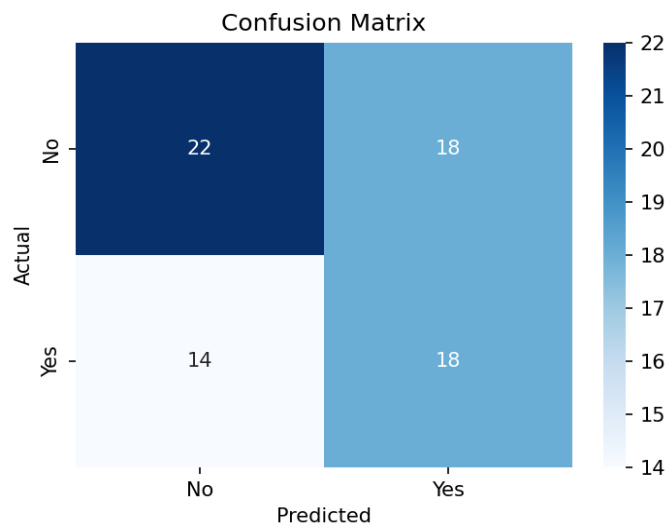
#### Training vs Testing Accuracy



#### Evaluation Metrics

| Metric            | Score  |
|-------------------|--------|
| Training Accuracy | 1.0000 |
| Testing Accuracy  | 0.5556 |
| Precision         | 0.5000 |
| Recall            | 0.5625 |
| F1-Score          | 0.5294 |

#### Confusion Matrix



|            | Predicted No        | Predicted Yes       |
|------------|---------------------|---------------------|
| Actual No  | 22 (True Negative)  | 18 (False Positive) |
| Actual Yes | 14 (False Negative) | 18 (True Positive)  |

## Overfitting Check

The Training Accuracy is 1.0000 and the Testing Accuracy is 0.5556. The gap between them is 0.4444, which is very large. This strongly indicates that the Decision Tree model is overfitting. The model has memorized the training data too closely and does not generalize well to new unseen invoices. A Decision Tree with no depth limit tends to overfit because it keeps splitting until every training record is correctly classified. To reduce overfitting, the tree depth could be limited using `max_depth` or `min_samples_leaf` parameters.

## Task 4: Business Interpretation of Model Results

### How well did the Decision Tree model perform?

The Decision Tree model achieved a testing accuracy of 55.56%. Out of 72 test records, the model correctly predicted 40 cases. The precision is 0.5000, recall is 0.5625, and F1-score is 0.5294. The model performance is poor and close to random guessing. This suggests the model is heavily overfitting and is not reliable enough for business use in its current form.

### Is there a large difference between training accuracy and testing accuracy?

Yes, there is a very large difference. The training accuracy is 1.0000 which is 100% and the testing accuracy is 0.5556 which is 55.56%. The gap between them is 0.4444, which is extremely significant. This means the model learned the training data perfectly but failed to generalize to new data, which is a clear sign of severe overfitting.

### Does the model show signs of overfitting? Why or why not?

Yes, the model shows very clear signs of overfitting. The training accuracy is 100%, meaning the model perfectly learned every record in the training data. However, the testing accuracy dropped to only 55.56% on unseen invoice data. This large gap of 44.44% means the model memorized the training patterns instead of learning general rules. A Decision Tree with no depth limit always overfits because it keeps creating branches until every training record is classified correctly. Applying `max_depth` or `min_samples_leaf` constraints is strongly recommended.

### Which evaluation metric is most important for this business problem?

Recall is the most important metric for PayFlow. A low recall means the model is missing invoices that will truly be paid late, which means PayFlow loses the opportunity to intervene early. Missing a late invoice (false negative) leads to cash-flow problems, delayed follow-up, and higher collection effort later. The current recall of 0.5625 means the model only correctly identified about 56% of invoices that were actually late. F1-score is also useful because it balances recall with precision, ensuring the company does not over-contact customers unnecessarily.

### What do false positives mean in this business context?

There are 18 false positives in the confusion matrix. This means the model predicted that 18 invoices would be paid late, but they were actually paid on time. In business terms, PayFlow would send unnecessary reminders or assign staff attention to customers who did not need it. This may create extra workload for collections staff and could annoy reliable customers if the communication is too aggressive, potentially damaging business relationships.

### What do false negatives mean in this business context?

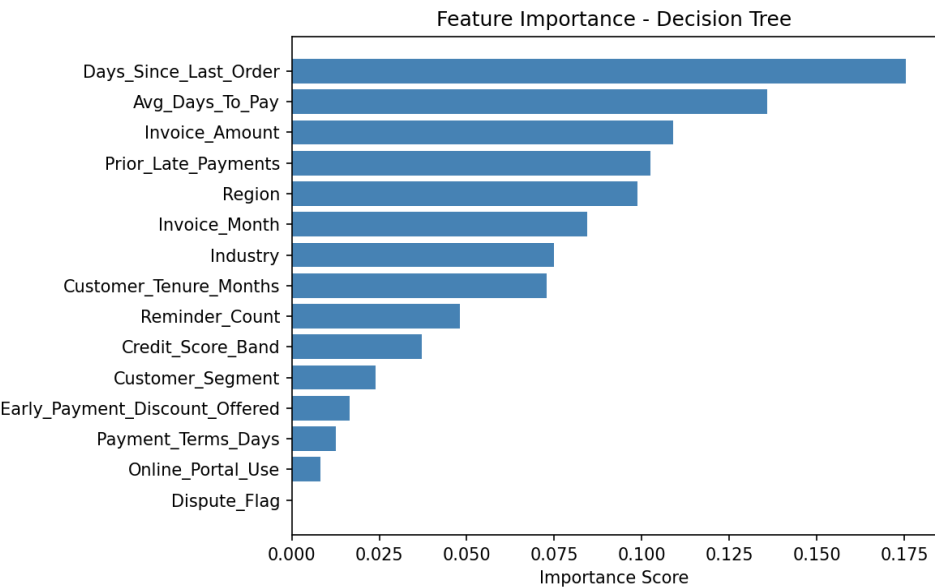
There are 14 false negatives in the confusion matrix. This means the model predicted that 14 invoices would be paid on time, but they were actually paid late. This is the more costly error for PayFlow because the company misses the opportunity to intervene early. These missed late invoices lead to cash-flow problems, delayed follow-up, and higher

collection effort later. For a financial operations company, false negatives directly impact revenue and customer account health.

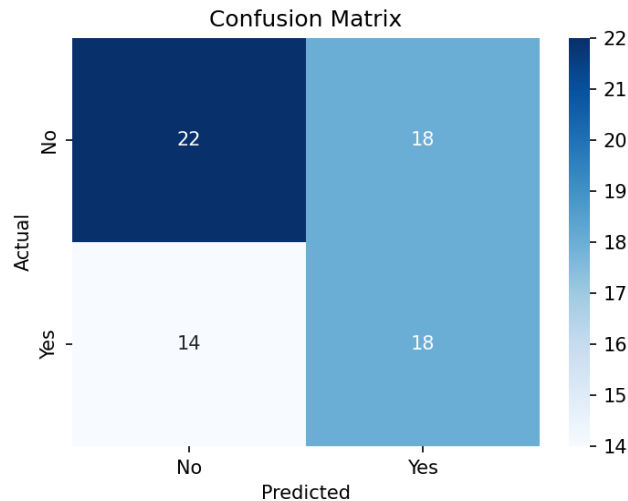
Which features were most important in the Decision Tree model?

| Feature               | Importance Score |
|-----------------------|------------------|
| Days_Since_Last_Order | 0.1754           |
| Avg_Days_To_Pay       | 0.1358           |
| Invoice_Amount        | 0.1090           |
| Prior_Late_Payments   | 0.1025           |
| Region                | 0.0989           |

Feature Importance Chart



Explanation of Metrics with Confusion Matrix



**Accuracy (0.5556):** The model correctly predicted only 55.56% of all invoices. Out of 72 test records, only 40 were predicted correctly. This is close to random guessing and indicates the model is not reliable in its current form.

**Precision (0.5000):** When the model predicted an invoice would be paid late, it was correct only 50% of the time. There were 18 false positives, meaning 18 unnecessary follow-up actions may have been triggered for customers who would have paid on time.

**Recall (0.5625):** The model correctly identified 56.25% of invoices that were actually paid late. Out of 32 genuinely late invoices, the model caught 18 and missed 14. The 14 false negatives represent invoices where PayFlow lost the chance to intervene early.

**F1-Score (0.5294):** The F1-score of 0.5294 is low, reflecting a poor balance between precision and recall. Both metrics are weak, which means the model needs significant improvement before it can be used to make reliable business decisions.

### How can these important features help the business make better decisions?

Days\_Since\_Last\_Order is the most important feature, suggesting that customers who have not ordered recently may be less financially engaged and more likely to pay late. PayFlow can flag invoices for customers with long gaps since their last order. Avg\_Days\_To\_Pay reflects historical payment speed and is a strong indicator of future behaviour. Customers who have historically taken longer to pay should receive earlier reminders. Invoice\_Amount suggests that larger invoices may carry higher late payment risk and may need dedicated account manager attention. Prior\_Late\_Payments directly signals high-risk customers with a history of late behaviour. Region indicates that geographic location plays a role in payment patterns and may reflect local economic conditions.

### What is one possible limitation or bias in the model or dataset?

The dataset is synthetic and does not reflect real-world payment complexity. One possible bias is that the model may unfairly associate certain regions or industries with late payment without enough context about economic conditions or contract terms. Real payment behaviour is also affected by factors not included in the dataset such as seasonal business cycles, customer financial stress, invoice approval workflows, personal relationships with account managers, and broader economic conditions. These missing variables reduce the reliability of the model's predictions.

### Why should human judgment still be used when making business decisions based on model results?

The model should act as an early warning system, not a final decision-maker. Given the current poor testing accuracy of 55.56%, human review is especially important. Staff should review high-risk predictions before taking action, particularly for large accounts, long-term customers, or invoices with active disputes. Human review can prevent unfair treatment, protect customer relationships, and identify cases where the data does not tell the full story. A responsible workflow would use the model to flag invoices for review, but employees should decide the best action based on customer history, communication records, and business context.

## Business Analysis

### SWOT Analysis

The SWOT analysis below evaluates the strengths, weaknesses, opportunities, and threats of using the Decision Tree model for PayFlow late invoice payment prediction.

|                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Strengths</b> <ul style="list-style-type: none"><li>• Explainable rules that finance teams can understand</li><li>• Automates late payment risk flagging across all invoices</li><li>• Avg_Days_To_Pay and Prior_Late_Payments are strong predictors</li><li>• Reduces manual follow-up workload for collections staff</li><li>• Fast and low-cost predictions for each invoice</li></ul> | <b>Weaknesses</b> <ul style="list-style-type: none"><li>• Model is overfitting with training accuracy of 100%</li><li>• Testing accuracy of 55.56% is low and unreliable</li><li>• Recall of 0.5625 means many late payments are missed</li><li>• Decision Tree is sensitive to small changes in invoice data</li><li>• Synthetic dataset may not reflect real payment behaviour</li></ul>                            |
| <b>Opportunities</b> <ul style="list-style-type: none"><li>• Apply max_depth pruning to reduce overfitting</li><li>• Add real-world data such as economic conditions and contract terms</li><li>• Integrate model into PayFlow invoicing dashboard</li><li>• Use predictions to trigger automated early reminders</li><li>• Expand model to predict expected days to payment</li></ul>       | <b>Threats</b> <ul style="list-style-type: none"><li>• False negatives cause missed late invoices and cash-flow problems</li><li>• False positives cause unnecessary customer contact</li><li>• Model may become outdated as payment behaviour changes</li><li>• Over-reliance on model without human review is risky</li><li>• Bias may unfairly associate certain industries or regions with late payment</li></ul> |

### Model Analysis

The Decision Tree Classifier was selected for this problem because it produces explainable and readable rules that finance teams can understand without technical knowledge. The model was trained on 288 records and tested on 72 records after an 80/20 stratified split.

| Aspect            | Detail                   |
|-------------------|--------------------------|
| Model Type        | Decision Tree Classifier |
| Training Set Size | 288 records (80%)        |
| Testing Set Size  | 72 records (20%)         |
| Training Accuracy | 1.0000 (100%)            |
| Testing Accuracy  | 0.5556 (55.56%)          |
| Precision         | 0.5000                   |
| Recall            | 0.5625                   |
| F1-Score          | 0.5294                   |



|                          |                                  |
|--------------------------|----------------------------------|
| Overfitting Gap          | 0.4444 (Very large gap detected) |
| Most Important Feature   | Days Since Last Order (0.1754)   |
| Second Important Feature | Avg_Days_To_Pay (0.1358)         |
| Third Important Feature  | Invoice_Amount (0.1090)          |

The model achieved a testing accuracy of only 55.56%, which is close to random guessing. The recall score of 0.5625 means the model missed 14 out of 32 invoices that were genuinely late. For PayFlow, these missed predictions represent lost opportunities to intervene early and prevent cash-flow problems. The overfitting gap of 0.4444 is very large, indicating the model must be significantly improved before business deployment. Applying max\_depth pruning, adjusting class weights, and incorporating more real-world features are the recommended next steps.

## Recommendations

Based on the model results and business analysis, the following recommendations are made for PayFlow to improve late payment prediction.

| Recommendation                                                                                   | Business Impact                                                               |
|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Apply max_depth pruning to reduce the overfitting gap of 0.4444.                                 | Improves testing accuracy and reliability.                                    |
| Improve Recall using class_weight to prioritize the Late Payment = Yes class.                    | Reduces false negatives and flags more late invoices early.                   |
| Set automated reminders for customers with high Avg_Days_To_Pay and Prior_Late_Payments.         | Enables proactive collection before invoices become overdue.                  |
| Include real-world data such as seasonal cycles and economic indicators.                         | Makes predictions more relevant to actual payment behaviour.                  |
| Use the model as an early warning system with finance staff reviewing all high-risk predictions. | Combines data insights with human judgment to protect customer relationships. |
| Retrain the model regularly as new invoice data becomes available.                               | Prevents model drift and keeps predictions aligned with current patterns.     |

## Overall Recommendation

The Decision Tree model in its current state is not suitable for business deployment due to severe overfitting and a testing accuracy of only 55.56%. However, the features identified as most important, particularly Days\_Since\_Last\_Order, Avg\_Days\_To\_Pay, Invoice\_Amount, and Prior\_Late\_Payments, provide valuable insights that PayFlow can act on immediately without relying on the model. The immediate priority is to apply tree pruning, improve recall, and test the model on real invoice data. Once improved, the model can be integrated into PayFlow's invoicing dashboard to flag high-risk invoices daily, allowing finance teams to focus their collection efforts where they matter most. Combined with human oversight and regular retraining, this approach will help PayFlow reduce cash-flow risk, lower collection costs, and support healthier customer payment relationships.